Climate inequality

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Abstract

This paper characterizes the relationship between anthropogenic climate change and economic inequality within countries, a subject intersecting two of the defining social challenges of the 21st century but which remains remarkably understudied. To address this urgent research gap, I build on cutting-edge methods for estimating climate impacts and apply these refinements to newly available distributional national accounts data. I document new evidence that temperature shocks significantly impacts income inequality within countries, an effect driven by disproportionate reductions in income among the poorest segments of already-poor countries and a surprising relative exposure of top quantiles to environmental shocks. The incidence pattern which is recovered amounts to a slightly progressive redistribution of a strictly smaller sum of income. To better understand the welfare implications of this incidence pattern in the context of systematic climate change, I integrate these reduced-form results over observed income distributions and use output from the newest generation of global climate models to perform a counterfactual analysis attributing historically observed economic inequalities to anthropogenic factors. These findings represent the most comprehensive characterization yet of the impact of climate change on the global distribution of income.

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1. Introduction

Mitigating anthropogenic climate change and reducing economic inequality are widely recognized two of the defining social issues facing the world in the 21st century. They are foundational to the 2030 Agenda for Sustainable Development adopted by all United Nations members, whose Sustainable Development Goals explicitly resolve to "take urgent action to combat climate change and its impacts" and "reduce inequality within and among countries" (United Nations, 2015). These are understood to be intimately related objectives: The Intergovernmental Panel on Climate Change identifies "widening opportunities between and within countries" (Masson-Delmotte et al., eds, 2018) as first-order considerations in any sufficient policy response to climate change and the World Inequality Report remarks that "climate mitigation is largely a distributional issue, not only between countries but also within them" (Chancel et al., 2022). Appreciation of these linkages extends to those who directly study the social impacts of climate change; the overwhelming majority of published environmental economists surveyed in 2021 indicated a belief that climate change is "likely" or "very likely" to exacerbate inequalities between (89%) and within (71%) countries (Howard and Sylvan, 2021).

Spurred by methodological advances, accessibility of computing power, and rising political demand, the past decade has seen a proliferation of empirical research dedicated to estimating the social impact of climate change. Contributions to this literature generally consider some form of a "damage function" which causally relates variation in weather variables taken to represent the global climate to variation in a social outcome of interest. Results are translated into an interpretable and usually monetarily denominated object such as a carbon price in order to inform the design of climate policy. Individual entries vary substantially in their implementations and are often split into broad "top-down" and "bottom-up" categories based on their empirical design. Top-down approaches restrict attention to a widely available outcome that is taken to be a comprehensive enough measure of social welfare that climatically attributable impacts to the stock of this outcome may represent the socially relevant "damage". Alternatively, bottom-up approaches consider impacts to distinct "sectors" of an economy; mortality, energy consumption, and agriculture are prominent examples. Estimates resulting from this approach can be presented as partial climate impacts or combined with other sectoral damages in an integrated model which accounts for their interdependencies and covariations to produce a more complete social cost estimate. The advantage of the former approach is in its convenience and interpretability: the wide availability and international comparability of economic aggregates

such as the production measures provided in national accounts data substantially simplifies the task of calculating global climate impacts. The benefit of the latter approach is in its potential to account for disaggregated, distributional, and non-market impacts to which single-dimensional aggregates are mostly indifferent.

Conversely, the disadvantages of the two approaches jointly constrain the economic analysis of climate change; the single-dimensionality of the top-down approach and prohibitive data demands of the bottom-up approach imply a critical tradeoff between geographic disaggregation and global coverage with severe consequences for our understanding of the relationship between climate change and global inequality. In perhaps the most prominent existing work directly concerned with characterizing this relationship, Diffenbaugh and Burke (2019) estimate that the ratio between the top and bottom deciles of the global income distribution was 25% larger in 2010 than it would have been had anthropogenic contributions to climate change been constrained to 1960 levels, all else held equal. But this is explicitly only a between-country measure of inequality; calculations result from an analysis which assigns all individuals their country's mean GDPs and defines deciles by relative national population sizes, an unavoidable restriction when limited to global data with only 160 or so distinct economic units. The authors remark that "documenting the impact of global warming on within-country inequality remains an important challenge."

Hsiang et al. (2017) is a particularly impressive example of a rare study which makes the opposite negotation between specificity and breadth in order to make progress in this area. Integrating fine-scaled agricultural, crime, energy, mortality, and labor data, the authors determine that climate risks in the continental United States are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts; climate change then is likely to imply worsening inequality within the United States. But to make more generalizable claims about the inequality implications of global warming, analyses must extend to settings which lack comparably rich or granular data.

Consequently, there is a critical dearth of evidence on the economic incidence of climate change across the eight billion people inhabiting the Earth beyond what is explained by differences across fewer than 200 national boundaries. In fact, Chancel et al. (2022) estimates that between-country differences have comprised a rapidly diminishing share of global income inequality since the emergence of the modern international order in the second half of the 20th century. By the group-decomposable Theil measure, between-country differences now account for just over 30% of global income inequality, down from

57% as recently as 1980. Thus, an understanding of global 'climate inequality' which is limited to differences across countries is not only unsatisfyingly agnostic to inequalities between people, but also half as informative as it was just 40 years ago and trending towards increasing irrelevance for the remainder of the 21st century. Progress towards understanding these within-country disparities may also improve prospects for an effective response to the climate change problem since democratic pressures and a richer portfolio of policy instruments can be more effectively mobilized in a domestic context than coordinated across a Westphalian international political economy (Barrett, 2003; Levin et al., 2009; Keohane and Victor, 2011).

This paper aims to make and inspire progress on this critical research gap by providing the most comprehensive characterization yet of the effects of climate change on global income inequality, placing particular emphasis on its underexamined within-country features. Achieving this requires overcoming several challenges which have historically inhibited progress in this area. First, prevailing empirical methods commonly used to estimate climate impacts require refinement. Even in standard country-level analyses using the same sources of data, estimated impacts of climate change are known to diverge by orders of magnitude depending on how the modeler chooses to specify the "damage function" causally relating weather to economic outcomes of interest (Pindyck, 2013). To avoid transmitting this imprecision to our new research setting, I extend cutting-edge impact models to incorporate features addressing outstanding methodological shortcomings flagged in recent literature. This methodological contribution is presented in Section 2.1 using an expositional application to conventional aggregate impacts for comparison to existing literature. In Section 2.1.2, I adapt new methods from empirical macroeconomics (Inoue et al., 2024) to provide direct statistical tests of the persistence dynamics whose unfalsifiability has largely driven the extreme sensitivity of cumulative impacts reported in the literature (Newell et al., 2021).

I then apply this refined apparatus to the most recent update to the World Inequality Database which provides annual country-percentile-level estimates of income harmonized for cross-country comparability and with expanded country coverage now representing almost the entire global population. This shifts the frontier of the specificity-breadth tradeoff described above by strictly disaggregating the analysis of global economic inequality. Section 2.3 documents new empirical evidence demonstrating that the impacts of identified temperature shocks on within-country inequality are substantial and driven primarily by disproportionately negative and relatively persistent impacts to lower income quantiles in already-poor countries. An asymmetrical hump-

shaped incidence pattern emerges which implies a net transfer of national income shares towards the middles of national income distributions, depicting a substantial regressive transfer from bottom quantiles and a proportionally smaller progressive transfer away from top quantiles. This latter feature measures a surprising degree of exposure of top incomes to environmental shocks relative to the middle classes which has not been previously observed to the best of my knowledge. Summing these impacts to the country level reproduces the standard result of likely net aggregate loss most pronounced in warmer countries, implying these inequality effects amount to redistributions of shares in a reduced total stock of income.

In Section 3, we integrate this reduced-form relationships over historically observed income distributions and weather to measure the influence of temperature shocks on the global distribution of income since 1980, the first year for which we have distributional data with near-global coverage. First, Section 3.1 provides estimates the cumulative impacts of variation in weather over this period to the global distribution of income. In Section 3.3, these effects are decomposed into natural and anthropogenic contributions using simulation output from state-of-the-art global climate models and methods developed in climate attribution science. A final decomposition in Section 3.3 splits these inequality effects into between-country and within-country dimensions. Together, the results presented in this section imply that anthropogenic climate forcings have almost certainty lowered the level of income across the entire global income distribution since 1980 with redistributive impacts more pronounced within countries than between them.

Section 4 concludes by considering how these new findings complement and complicate existing aggregate-level evidence on global climate impacts and suggesting how they may inform the design of climate policy. I close by considering the limitations of the present study design and propose promising lines of future research recognizing distributional welfare impacts as a first-order consideration in the economic analysis of climate change.

1.1 Related literature

The distributional focus of this paper complements and draws from a vast empirical literature estimating the economic "damages" of climate change, traditionally evaluated at sectoral, national, or global levels. Although long-term economic impacts implied by the models used in this literature are notoriously inconsistent (Newell et al., 2021), there is general agreement across models regarding the presence of non-linearities in the response of aggregates such as GDP to variation in temperature wherein marginal impacts increase

with a country's baseline temperature. These features motivate the methodological refinements described in Section 2 to address the extreme divergence in dynamic effects while maintaining the state-dependency broadly accepted by the literature. These non-linearities imply a country-level incidence pattern wherein climate damage and risk is concentrated in generally poorer countries located closer to the tropics, exacerbating between-country economic inequality, by now a widely accepted feature of climate change (Howard and Sylvan, 2021). A subset of these studies specifically detail the implications of this between-country dimension of inequality directly, such as Diffenbaugh and Burke (2019), described earlier. A related strand of between-country inequality studies highlight the inverse relationship between contributions to cumulative global greenhouse gas emissions and their exposure to their social consequences (Chancel and Piketty, 2015; Ricke et al., 2018; Kotz et al., 2024).

Similar non-linearities emerge at the micro level across a range of outcomes as varied as labor supply (Graff Zivin and Neidell, 2014), agricultural yields (Schlenker and Roberts, 2009), standardized test performance (Goodman et al., 2020), even social media profanity (Baylis et al., 2018). Carleton and Hsiang (2016) overviews the proliferation of multidisciplinary studies in this category whose results are generally suggestive of patterns of increased inequality within countries similar to those observed in the aggregate. The present study bridges this micro evidence with the aforementioned, estimating distributional responses consistent with the general features of both categories of impacts.

The literature on climate impacts on economic inequality *within* countries remains comparatively sparse in large part due to an absence of adequate research infrastructure. Estimation of distributional equivalents to the aggregate damage functions standard in the literature requires data comparable in credibility and coverage to domestic production data whose construction across countries follows rigorous standards set forth by the international System of National Accounts (European Commission et al., 2008). Historically, distributional measurement of welfare has lacked and arguably been crowded out by the wider interest and investment in measuring aggregate production (Jorgenson, 2018). While secondary inequality data products compiling country-year inequality variables from a wide array of official and survey sources date at least as far back as Deininger and Squire (1996), the validity of these products for cross-country causal analysis have long been known to be limited by inconsistencies across primary sources in construction and availability (Atkinson and Brandolini, 2001). For example, the most recent rigorous evaluation I could find of these secondary datasets concluded that

results derived from the conveniently packaged Standardized World Income Inequality Database should be regarded as "not sufficiently credible" and that analyses using the United Nations' long-running World Income Inequality Database (UNU-WIDER, 2021) should at best be accompanied by justifications for their sample and variable selection criteria (Jenkins, 2015). Nonetheless, these inequality products have been used in at least two recent studies (Cevik and Jalles, 2023; Gilli et al., 2024) relating the inequality measures provided in these products to variations in weather.

In the absence of credible global inequality measures, a handful of works such as Hsiang et al. (2017), which was described earlier, have managed to contribute to the study of within-country inequality impacts by using idiosyncratic data available for individual countries. Marx (2024) uses rich tax data from France to infer that years with marginally more days above 30°C are associated with wider income disparities between the richest and poorest cantons (administrative units described as roughly equivalent to 10 municipalities). Dasgupta et al. (2023) uses data from perhaps the most unequal country in the world to project a 3-6 Gini point increase in South Africa by 2100 under a moderate warming scenario.

The research programs of Thomas Piketty, Emmanuel Saez, and their coauthors since the turn of the century are widely credited with revitalizing public and academic interest in the measurement of within-country inequality. Beginning with work tracing the long-run evolution of income and wealth inequality in France (Piketty, 2003) and the United States (Piketty and Saez, 2003), their mobilization of extensive tax returns data has enabled credible measurement of top incomes over long time horizons, stimulating new programs of research and equipping the general public with new political language around which to mobilize (Jones, 2015). This paper takes advantage of the cross-country research apparatus that has rapidly developed and been made accessible in recent years with the intention of maturing the economic analysis of climate change in a similar manner. Palagi et al. (2022) is the only other climate-economic study I could locate which makes use of the same "distributional national accounts" (World Inequality Lab, 2024) used in this study. The authors find that extreme precipitation negatively impacts low-income individuals much more severely in countries more dependent on agriculture.

Finally, I locate this paper in the context of a growing demand for distributional considerations in economic analyses of climate change and the design of climate policy. For example, Chancel et al. (2023) estimates that within-country inequality of carbon emissions have recently exceeded the equivalent "carbon inequality" between countries and advocates for targeted carbon tax-

ation programs based on individual footprints to more efficiently internalize the climate change externality. The modern SCC framework accommodates equity considerations by concavifying the underlying utility function so that the welfare weights assigned to individuals are inversely related to their relative levels of consumption; a logarithmic specification is commonly chosen, for example. Variations of this progressive utilitarianism have been institutionalized to some degree in a handful of countries including recently in the United States (Office of Management and Budget, 2023). Theoretically, the way equity weighting is implemented should result in substantially higher SCC estimates depending on the extent to which the distributive effect of climate change is modeled as regressive[1]. This paper provides the first credible estimates of this regressivity at a near-global subnational resolution.

2. Empirical analysis

2.1 Empirical methods

2.1.1 Identifying temperature shocks

Until recently, it has been customary in the empirical climate impacts literature to use raw levels of weather variables such as temperature or rainfall as primary explanatory variables of interest. This convention is motivated by the intuition that annual fluctuations in temperature are plausibly exogenous with respect to most economic activity. Recent work such as Kahn et al. (2021); Bilal and Känzig (2024); Nath et al. (2024) challenge this convention on the grounds that expanded production correlates with weather through its association with greater emissions of greenhouse gases and that temperature levels exhibit substantial autocorrelation which confounds estimation. In place of raw temperature, they each propose different constructions of a temperature *shock* intended to better represent adaptive expectations and satisfy exogeneity conditions for valid dynamic inference.

Ramey (2016) outlines these conditions for general time series settings. For a candidate treatment variable to represent a valid shock, it must correspond to unanticipated movements in exogenous variables, be exogenous with respect to current and lagged endogenous variables, and be uncorrelated with other exogenous shocks included in the regression model. We follow the construction of Kahn et al. (2021) in defining shocks $\widehat{\tau}_{it}$ with respect to an M-period moving average of local temperature:

[1]Prest et al. (2024) show that one proposed implementation would increase the US government's official SCC estimate by a factor of eight

$$\overline{T}_{it} := \frac{1}{M}\sum_{m=1}^{M} T_{i,t-m} \tag{1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{1b}$$

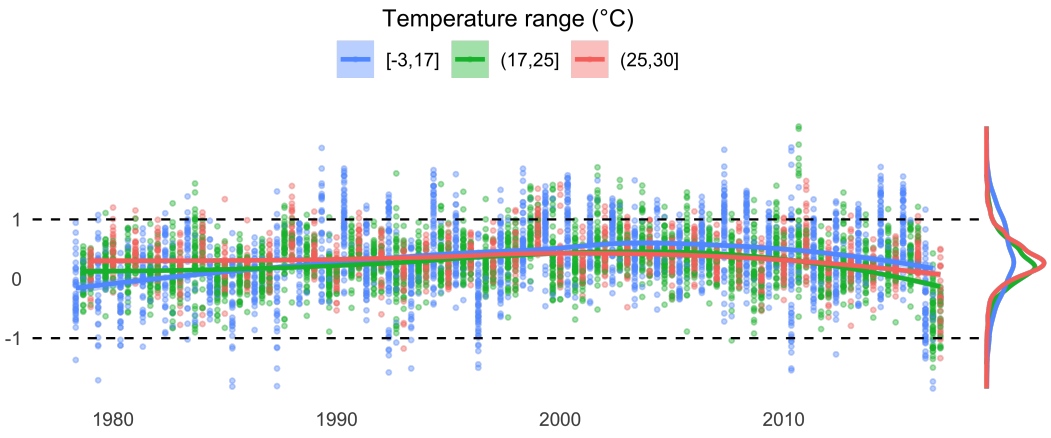
Figure 1 depicts country time series of the resulting temperature shocks with colors and flexible local regression curves corresponding to an even split of all country-years into three groups defined by their average temperature. The positive values of the local regression curves over almost the entire observation period capture the idea that systematic warming is characterized by relatively frequent and large "hot" shocks compared to "cold" shocks even if expectations about weather adapt to recent weather. Intervals where the running average shock is closer to 0 correspond to relative stability in the (lagged) climate. The marginal distribution plots on the right demonstrate that colder seasonal countries experience much larger temperature shocks than their warm tropical counterparts, a reflection of both greater natural weather variability and the fact that anthropogenic warming is generally greater in colder regions furthest from the equator.

As with the other candidate formulations, shocks defined this way satisfy the exogeneity conditions when lags are included as controls in the regression model being estimated.

I consider the simple definition above to be the most compelling construction of a temperature shock for several reasons. First, for sufficiently large M, this definition of the state variable $\overline{T}_{it}$ has an appealing coherence with the climatological convention of using long-run moving averages of weather realizations to anchor the climate at a fixed point in time. Most commonly, these "climate normals" are defined as 30-year averages to account for the fact that even a stable climate may exhibit cyclical weather patterns spanning several years. The most economically relevant example of this natural variability is the El Niño Southern Oscillation, an ocean-warming phenomenon which can vary average global surface temperatures by as much as 0.4 across its three phase cycles which may span 2-7 years. Relatively minor oscillations can span multiple decades.

Secondly, these definitions of the shock and state variables are much more consistent with the empirical literature which generally reports "limited evidence" of adaptive behavior moderating the direct impacts of weather shocks in the context of climate change (Burke et al., 2024). Here, the state $\overline{T}_{it}$ is interpreted as the time-varying expected value of temperature, representing beliefs about the unobserved probability distribution from which the weather

Figure 1: **Country-level time series of identified temperature shocks**



Points correspond to shocks observed in individual country-years. The three colors correspond to an equal-sized binning of country-years by their moving-average temperature. Flexible local regression curves describe time trends split by this binning. Marginal distribution plots on the right show that colder country-years observe greater variance in shocks.

realized in country i at time t is drawn, i.e., the local climate. Since the climate at time t would be best represented by the climate normal centered on time t, which is a function of future realizations, there exists an $M/2$-period lag between the "true" climate and the expected climate $\overline{T}_{it}$. Over the span of our historical data, we find that this difference is non-trivial: the median country observes, on average, a 0.26°C (IQR: 0.18-0.36) increase in the 30-year moving average every 15 years.

This lag between true and expected climate can be interpreted as a bound on the ability for belief or anticipation effects to offset the direct effects of weather shocks (corresponding to climate adaptation in the sense of Kahn et al. (2021) and Hsiang et al. (2017)) or simply as inattentive beliefs. This gives rise to the increasingly positive trends observed in Figure 1 during periods of more rapid climate change: indeed, one of the main channels through which climate change imposes economic harms is through the increased frequency and magnitudes of positive "hot" shocks relative to negative "cold" shocks. To the extent that local expectations about the climate are anchored by recent experience, shocks defined this way may be interpreted as deviations relative to adaptive expectations; a given temperature level may register as a "hot" shock one year but a "cold" shock in the same country decades later if the intermediate period is characterized by systematically warmer weather that its inhabitants have internalized either through some concept of informational updating or since it constitutes the entire life experience of a new generation.

This contrasts with the constructions of Bilal and Känzig (2024) and Nath et al. (2024) (henceforth, NRK), which define shocks as residuals from autoregressive distributed lag models of temperature. In so doing, they implicitly impose a hyper-responsive model of adaptation wherein expectations are optimally set each period to minimize the direct economic effects of temperature shocks. As a result, the distribution of temperature shocks are shown to be stationary with zero mean even in the context of severe climate change. A compromise between these two models of adaptation can be accommodated in our construction by reducing M or imposing a weighted average scheme which places greater weight on more recent observations.

2.1.2 Estimating responses to a temperature shock

We adapt the semi-parametric local projections method pioneered by Jordà (2005) to estimate state-dependent impulse responses to a temperature shock. As Jordà (2023) notes, the method has found growing appeal outside its home field of applied macroeconomics as its flexibility (Cloyne et al., 2023), relevance to the potential outcomes framework (Dube et al., 2024), and robustness

to misspecification and non-stationarity (Montiel Olea and Plagborg-Møller, 2021; Montiel Olea et al., 2024; Piger and Stockwell, 2023) have been established in recent econometric literature. Berg et al. (2024), Bilal and Känzig (2024), and NRK are other climate papers which apply the method to estimate dynamic economic responses to temperature shocks.

These features are particularly useful for addressing several outstanding identification concerns flagged in reviews of impact estimation methods. The remainder of this section uses an application to country-level GDP impacts to aid exposition of the methods used in this study which will also serves to familiarize the reader with features of the aggregate impacts literatures upon which the inequality results build.

We use panel data to estimate dynamic responses to temperature for several distributional outcomes. For now, we take y_{it} to represent the logarithm of GDP for country i in year t. The local projections method entails estimating $H+1$ iterations of the following single-equation model corresponding to different projection horizons $h \in \{0, 1, ..., H\}$:

$$\begin{aligned} \Delta_h y_{i,t+h} &:= y_{i,t+h} - y_{i,t-1} \\ &= \beta_{1h}\widehat{\tau}_{it} + \beta_{2h}\widehat{\tau}_{it} \cdot \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + u_{i,t+h} \end{aligned} \tag{2}$$

The matrix $\mathbf{Z}_{it}$ includes lagged controls $\{\widehat{\tau}_{i,t-j}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta y_{i,t-j}\}_{j=1}^{p}$. The inclusion of p lags of the shock $\widehat{\tau}$ accounts for potential autocorrelation in the treatment, validating its use as a shock as described in the previous subsection. Interactions of these lagged shocks with the state variable capture potential state-dependencies in these autocorrelations. Lags of the outcome are included as a guard against trend non-stationarity. Finally, the set μ_i and η_t of standard linear fixed effects are included to absorb time-invariant country-specific effects and common year-specific effects. Besides the construction of the shock and permitting the state variable to vary over time, this model borrows directly from the static state-dependent model of NRK.

Each projection horizon h thus attributes a main effect β_{1h} representing the impact of a unit pulse of temperature on the "long difference" of y projected h periods after the shock. An interaction term allows the effect to scale linearly with the country-year's long-run average temperature $\overline{T}_{it}$. This state-dependency is motivated by the standard result in the climate impacts literature that the economic effect of a temperature shock is increasingly negative for warmer climates so we should expect this coefficient to be negative. Finally, the inclusion of a main effect λ_h associated with the time-varying state variable allows for the possibility that a systematic change in the climate even

within the same country may impact expected growth independent of its impact on the distribution of shocks. For example, if a hot country stabilizes at a climate that is 1°C warmer after an intermediate period of warming, it is plausible that economic productivity may be systematically lower even when given sufficient time to maximally adapt. This will not affect the estimation of an impulse response to a shock of temperature but will be relevant for the historical analysis in Section 3.

It is worth taking stock of the restrictions imposed by the model described in (2). The flexibility of this semi-parametric approach is particularly appealing in the context of a literature where the magnitudes of estimated impacts are known to be largely driven by the specification of the damage function (Newell et al., 2021; Chang et al., 2023). For example, two studies projecting the proportional loss in cumulative global production by 2100 arrive at point estimates of -3.4% (Casey et al., 2023) and -20% (Burke et al., 2015) under the same worst-case warming scenario with almost all this divergence stemming from the decision to specify the outcome in levels (e.g., log GDP) or as growth (e.g., first differences in log GDP). Section A1.1 and Figure A1 describe how this choice amounts to assuming outright that temperature has a one-off or permanent effect and how alternative persistence structures where level effects are temporary but persistent can be accommodated by either model by including p additional lags of the explanatory variables.

However, as the persistence being considered increases in horizon, the efficiency gains from parametric specifications are rapidly diminished to accommodate increasingly imprecisely estimated dynamic effects. The efficacy of lag augmentation is further constrained by non-linear functional forms and autocorrelation in the explanatory variables, both general features of damage functions where absolute temperature is used as an explanatory variable. Indeed, Burke et al. (2015) conclude that the identifying variation which remains after including lags "cannot reject the hypothesis that this effect is a true growth effect nor the hypothesis that it is a temporary level effect", rendering these extremely influential parametric choices unfalsifiable.

By comparison, the local projections method imposes no restriction on the evolution of a given IRF over time since the effects associated with each projection horizon are estimated from entirely separate regressions and different subsamples[2] In addition, the local projection specified in long differences has been shown to be particularly robust to biases associated with non-stationarity processes Piger and Stockwell (2023); Jordà (2024) such as those which plague standard GDP regressions specified in levels (Nelson and Plosser,

[2]Residuals across the projection horizon are generally autocorrelated but do not affect consistency of point estimates.

1982; Campbell and Mankiw, 1987). Altogether, the method is relatively robust to different forms of misspecification by remaining mostly agnostic about the underlying data-generating process.

Collecting estimates of the coefficients of interest $\{\widehat{\beta}_{1h}, \widehat{\beta}_{2h}\}_{h=0}^{H}$ and adopting the notation of Jordà (2023), a set of local projection estimates imply a family $\mathcal{R}_{\tau \to y}(h; \overline{T})$ of state-dependent impulse responses as a function of the state variable $\overline{T}$:

$$\mathcal{R}_{\tau \to y}(h; \overline{T}) := \widehat{\beta}_{1h} + \widehat{\beta}_{2h} \cdot \overline{T} \tag{3}$$

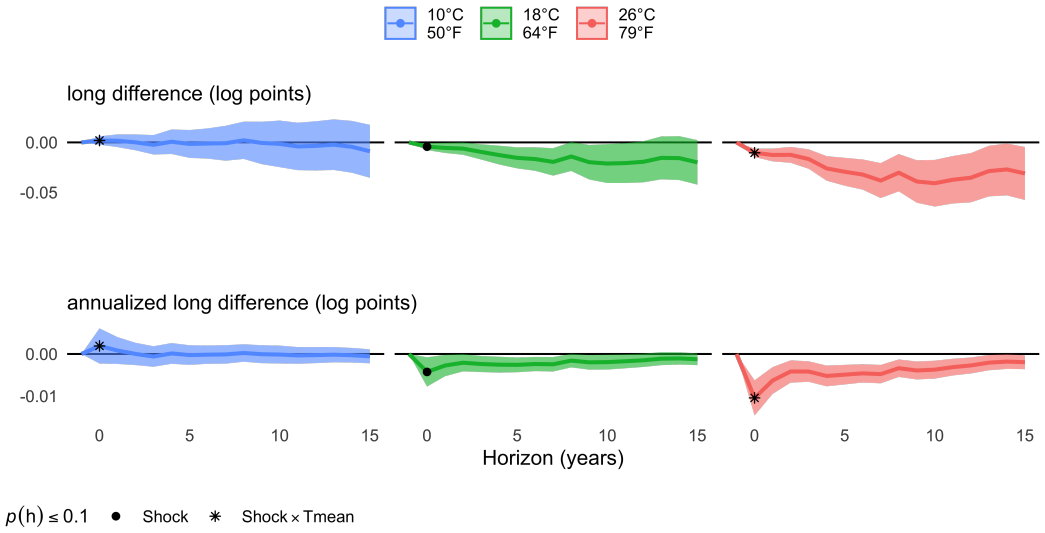
For example, the top row of Figure 2 depicts three different IRFs which each result from the same regressions but are distinguished by different assumed values of the state variable. These representative states correspond to the 20th and 80th percentiles of country-year temperatures in our sample (10°C colored blue and 26°C colored red) and their average (18°C, roughly the 40th percentile and colored green). Note that for expositional convenience, we have transformed our state variable to center on the intermediate reference temperature of 18°C so that the responses in green correspond to the sequence of estimated main effects $\widehat{\beta}_{1h}$ while the responses in blue and red incorporate equivalent but opposite interaction effects corresponding to ±8°C deviations from this reference state so that point estimates are calculated as $\widehat{\beta}_{1h} \pm 8\widehat{\beta}_{2h}$[3]. This coloring scheme and these reference temperatures will be used throughout the rest of this paper.

Compared to a standard growth regression from which permanent level effects are inferred from observations contemporaneous to the shock, the LP method conservatively estimates persistent effects at each horizon h separately using observations of changes in outcome y over h periods and as depicted by the corresponding IRF $\mathcal{R}_{\tau \to y}(h; \overline{T})$. Leaving aside the question of valid inference for now, interpretation of a resulting IRF proceeds as follows

1. **Level effect**
    - One-period level effect: $\mathcal{R}_{\tau \to y}(h; \overline{T}) = 0 \ \forall \ h > 0$
    - Persistent level effect: $\exists \ h^* > 0$ such that $\mathcal{R}_{\tau \to y}(h; \overline{T}) = 0 \ \forall \ h > h^*$
2. **Growth effect**

[3]To be specific, we redefine the state variable as $\overline{T}_{it} := \frac{1}{30} \sum_{m=1}^{30} (T_{i,t-m} - 18)$. Under this transformation, the blue and red IRFs correspond to the states $\overline{T}_i - 8 = 10$°C and $\overline{T}_i + 8 = 26$°C respectively.

Figure 2: **GDP impulse response to a 1°C shock**



The plots in the top row correspond to estimates of impulse responses of GDP estimated by the $H+1$ local projections regressions specified in (2) for three representative temperatures depicted in different colors. The plots in the bottom row depict the same functions annualized by dividing estimates by $h+1$. All figures in this paper adopt the same color-temperature correspondence where relevant. Points in black correspond to point estimates deemed significant by the joint hypothesis test described in Section 2.1 with filled circles corresponding to significant main effects and asterisks corresponding to significant interaction effects.

- Simple growth effect: $\mathcal{R}_{\tau \to y}(h; \overline{T})$ converges to a non-zero constant
- Divergent growth effect: $\mathcal{R}_{\tau \to y}(h; \overline{T})$ never converges to a constant

The contemporaneous estimates in the IRFs in the first row of Figure 2 depict a pattern now standard in the aggregate impacts literature featuring near-null temperature effects in cold climates, mild effects in mid-temperature climates, and the largest negative effects in hot climates. The evolution of these impacts over time is less commonly depicted and here produce a pattern qualitatively similar to those reported in NRK wherein per capita GDP appears to stay persistently depressed over the entire projection horizon. The second row depicts the same IRFs annualized by scaling down estimates by their corresponding horizon $h+1$ to more clearly depict convergence behavior relative to the contemporaneous effect of the shock.

Of course, the standard bias-variance tradeoff applies so that the attractive flexibility features described above come at the cost of efficiency; IRFs estimated by local projections are generally noisier than their fully parametric and structural counterparts[4].

Importantly, the LP model imposes a linearity assumption on effects estimated within the same horizon. Since the shock $\widehat{\tau}_{it}$ and the interaction $\widehat{\tau}_{it} \cdot \overline{T}_{it}$ enter additively into (2), both effects are assumed to scale in proportion to the size of the shock so that a 1°C shock has exactly twice the effect of a 0.5° shock and the exact negative of the effect of a -1°C shock. In reality, non-linearities abound in the interactions between social and climate system (Dietz et al., 2021). Additionally, the linearity assumption imposes the same symmetry in the interaction term so that relative to the reference temperature, estimated interaction effects for states a given number of degrees warmer than the reference temperature will always be the exact negative of those for states the same number of degrees cooler. In Figure 2, this is reflected in the fact that for a given horizon h, the point estimates in green will always be the exact average of those in blue and red since they are estimated from the same linear regression model. Both of these assumptions can be relaxed in alternative specifications in which the shock and the interaction enter non-linearly, but we do not consider these in the current study.

It is worth noting that our empirical estimates are expressed in terms of responses to a 1°C shock, which is substantially larger than a shock we would expect to observe in a single year as we showed in Figure 1. The median ab-

[4]Different compromises in this tradeoff can be obtained by imposing additional restrictive parametric assumptions such as the functional approximation approach of Barnichon and Matthes (2018) which can substantially narrow the error bands.

solute value of a shock (whether positive or negative) varies by the rate of climate change which will in turn vary across states and time. For the warmest 33% of country-years, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37 in the ensuing decades and so hot-country effects can be understood as corresponding to a shock 3-5 times larger than one would expect to occur in those climates in a given year. For the coldest 33% of country-years, the median magnitude of a shock was about 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as just 2-3 times larger in magnitude than the median shocks.

Global inequality effects of climate change thus are not just dependent on the shape of the impulse response responses across quantiles but also the distribution of warming over space and time as well as the actual distributions of incomes across quantiles. The simulation exercises to be described in Section 3.1 are valuable because they combine four relevant climate inequalities: inequality of temperature effects across an income distribution, inequality of incidence of temperature effects across different climates, inequality in the global distribution of income, and inequality in the distribution of warming.

Inferring persistence from local projection estimates

The error bands depicted in Figure 2 reflect pointwise estimation imprecision. This estimation imprecision is particularly important to include in this context since the volume of regressions underlying an LP-IRF precludes convenient summarization in a standard regression table[5]. One may be tempted to infer from the fact that the hot-country bands in red do not intersect the horizontal axis over the entire projection horizon that these estimates constitute significant evidence to reject a null hypothesis of level effects which persist for less than 15 years.

However, the uncertainty depicted by error bands correspond to individual hypothesis tests. Since estimates and residuals across an IRF are serially correlated—for example, the green bands straddle the horizontal axis but all estimates are negative— the relevant test of significance is a joint hypothesis test not captured by pointwise error. Focusing on the impulse response corresponding to the main effect, the significance of the corresponding impulse response $\mathcal{R}_{\tau \to y}(h)$ amounts to testing the null hypothesis that all main effects are 0:

[5]All graphically depicted regressions presented in this paper correspond to a significance level of $\alpha = 0.1$

$$H_0(h) : \beta_{1,0} = \beta_{1,1} = ... = \beta_{1,h} = 0$$

Rejecting the null for at least one of the $h+1$ component main effects is sufficient to reject the null of the total impulse response up to horizon h. Since the number of coefficients being considered in the joint hypothesis increases in h, the coverage of the joint hypothesis test must be adjusted for the inclusion of each additional horizon to compensate for the problem of multiple hypothesis testing. Inoue et al. (2024) derive convenient estimators for the implied "significance band" defined to satisfy the following condition:

$$\mathbb{P}\left[\bigcap_{h=0}^{H}\left\{\zeta_{\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}} < \widehat{\beta}_h < \zeta_{1-\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}}\right\}\right] \geq 1-\alpha$$

The scaling of the significance level α by $2(H+1)$ is an implementation of the Bonferroni correction. Graphically, these bands are represented by a region straddling the horizontal axis such that estimates of β_{1h} contained within it are insufficiently large enough to reject the hypothesis of a null effect that the impulse response persists up to period h. The width of these bands increases with the projection horizon, corresponding to the fact that for each projection horizon, the available estimation sample is reduced by one period (in the case of a perfectly balanced panel).

These significance bands are specific to the main effect and are represented by the gray regions in the middle panels of Figure A2. Since ours is a state-dependent model, I derive equivalent significance bands specific to the interaction effects. These are defined relative to the point estimates of the main effects and are depicted by the gray regions in the panels on the left and right. Points outside these regions are considered sufficient to reject the null hypothesis that interaction effects do not persist up to h periods. By the linearity assumption, the null hypothesis of zero interaction effects for a 10° climate is rejected if and only if it is rejcted for a 26° climate.

For comparison, Figure A3 depicts a placebo test where I construct bands to implement significance tests equivalent to those described above to test the persistence in the effect of temperature on another widely available and potentially non-stationary growth rate: that of population. As I am not aware of any theory or evidence to suggest temperature shocks would meaningfully impact country-level population growth[6], we should expect the tests to re-

[6]While there is an established literature on climate impacts on mortality rates (Barreca and

turn precise null results. Indeed, we find with all point estimates resulting from two separate population datasets easily contained by the corresponding significance bands. Notably, if we were to misuse the error bands as indicators of statistical significance, we would misinterpret the estimated IRFs as evidence to reject the null for large subintervals of the projection horizon for mid-temperature and hot climates. Instead, the entirety of these error bands are easily enveloped by the significance bands.

Since including significance bands, color-coded error bands, and point estimates for multiple IRFs can clutter space, I adopt a convention of omitting the significance bands entirely and instead coloring black the point estimates which are located outside the implicit error bands. In all IRFs beginning with those depicted in Figure 2, rounded points correspond to significant main effects while asterisks correspond to significant interaction effects. By this measure, these results cannot reject the null hypothesis of one-period level effects.

2.1.3 The state-dependent cumulative multiplier

Another reason to prefer temperature shocks to levels of temperature as treatment variables is that models which use the latter neglect to sufficiently account for autocorrelation in temperature itself when estimating dynamic effects (Nath et al., 2024; Bilal and Känzig, 2024). An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be supplemented later by stimulus programs in later periods. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; multipliers scale these effects at each horizon by the corresponding accumulation of shocks.

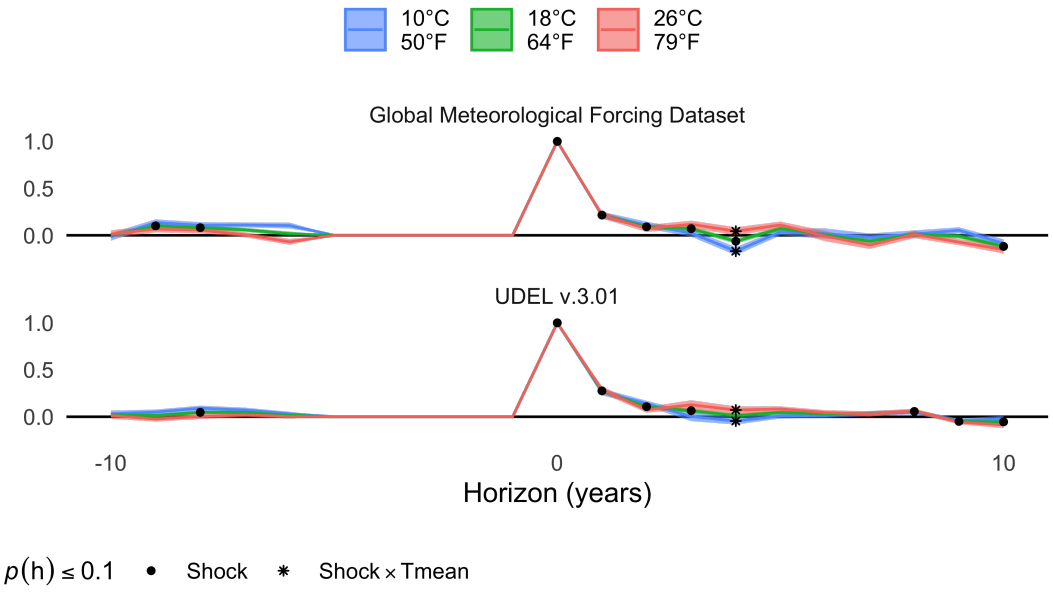
Figure 3 depicts the impulse response of temperature shocks to a pulse of itself for our three representative climates. Each exhibits very similar persistence patterns across two independent temperature dataset. An application of the persistence test covered in Section 2.1.2 implies autocorrelations which persist for up to 10 years though only the first 1-2 periods are qualitatively large (approximately 21% and 9% respectively with insignificant heterogeneity across climates).[7] Cumulative multipliers are estimated in order to scale

Shimshack, 2012; Barreca et al., 2016; Carleton et al., 2022), absolute magnitudes are miniscule relative to total populations

[7]This contrasts with the finding of NRK who measure shock persistence using levels of temperature to infer that temperature shocks remain 10-20% higher for at least nine years following a unit shock.

down the estimated impulse response of GDP $\mathcal{R}_{\tau\to y}(h)$ by the impulse response of the shock to itself $\mathcal{R}_{\tau\to\tau}(h)$ to account for this dynamic treatment schedule.

Figure 3: **Persistence of temperature shocks**



State-dependent impulse responses of identified temperature shocks estimated using two widely used temperature datasets each imply significant autocorrelation. In both cases, persistence is large in magnitude for the first 1-2 periods following a shock before flattening. Negative horizons are included as a check for pre-shock trends.

Cumulative responses are traditionally computed by first constructing *cumulative* versions of the two IRFs: $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to y}(j)$ for the outcome of interest y and $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to\tau}(j)$ for the shock $\widehat{\tau}$. Then the cumulative multiplier $m_{\tau\to y}(h)$ would be calculated as the ratio $\mathcal{R}^c_{\tau\to y}(h)/\mathcal{R}^c_{\tau\to\tau}(h)$. However, this approach has a couple of inconvenient downsides. First, as noted by Jordà (2024), computation of standard errors of a ratio of two random variables is complicated and non-standard. Second, it is computationally inefficient in the sense of requiring $2(h+1)$ (or $4(h+1)$ in our state-dependent

case) separate local projections to obtain the corresponding multiplier for each projection horizon h.

Instead, we follow the recommendation to use the one-step method described in Ramey (2016) and applied in Ramey and Zubairy (2018). We adapt the method for our state-dependent setting through the procedure summarized by the algorithm summarized in Algorithm A1.

We can use the resulting coefficients to trace a cumulative multiplier function $\mathcal{R}^{c}_{\tau\to y}(h;\overline{T})$ analogous to (3) but which accounts for the effect of persistence in the shock. Since estimation of the coefficients follows directly from a system of local projection regressions, standard inference is preserved and the significance testing methods outlined in Section 2.1.2 are equally available. Figure 4 demonstrates an application of the method to estimate temperature multipliers of GDP. The estimated multiplier over horizon h is interpreted as the difference in cumulative GDP over the h periods following an isolated 1°C shock relative to a counterfactual where the shock had never occurred.

Estimated cumulative multipliers are found to be substantial at larger horizons but the significance band test cannot reject a null of non-persistence beyond one period. 90% confidence intervals for the effect of an identified temperature shock on cumulative GDP are $[-0.49, 1.1]$ percent for 10°C climates and and $[-2.7, -1.1]$ percent for 26°C climates.

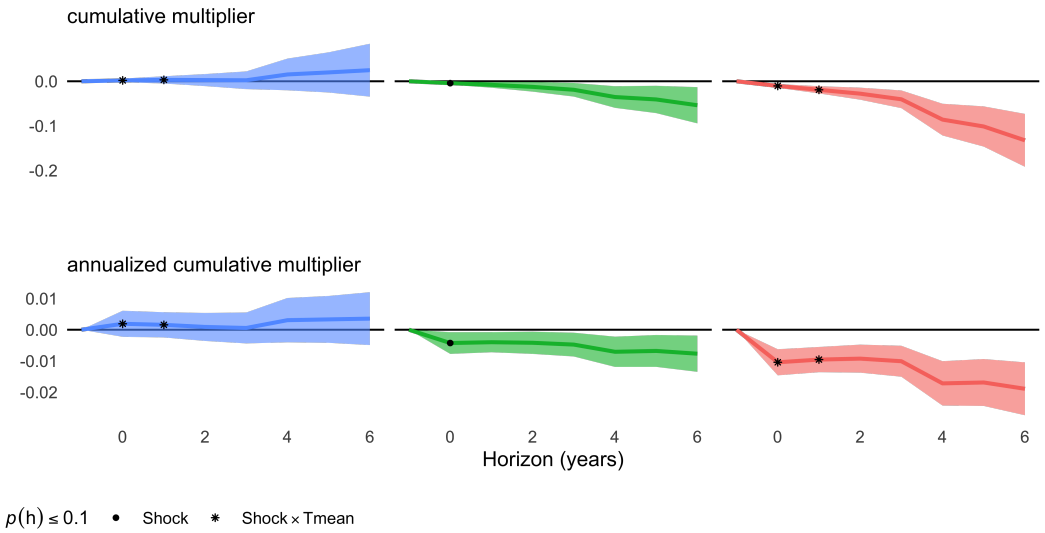
2.2 Data

In Section 2.1, we demonstrated how our empirical approach may complement and contribute to the vast existing literature on the aggregate economic impacts of weather shocks. We now apply the same techniques to a relatively unexplored setting, providing new evidence of the distributional effects of identified temperature shocks within countries and globally.

Income

This analysis is made possible by newly available grouped income data published by the World Inequality Database (World Inequality Lab, 2024), the successor to the tax data-based World Top Income Database which launched in 2011 with data on over 30 mostly Western countries. A critical difference between the two products was the development of a system of standardizing the integration of tax, national accounts, and survey data (Piketty et al., 2019) modeled after the System of National Accounts (European Commission et al., 2008). This newer generation of inequality data extends coverage within those countries—now claiming to capture "100% of income" in the United States

Figure 4: **Cumulative GDP impulse response to a 1°C shock**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

(Piketty et al., 2018)—and rapidly extending coverage and comparability to countries with relatively limited available data. By 2019, historical data for over 120 countries had been incorporated into the database. As of this writing, their equal-split adult pre-tax income series was available for 215 countries and territories collectively representing almost the entire global population with distributional data imputed every year since 1980. All incomes are converted to 2023 US dollars using the provided PPP-adjusted exchange rates.

Accompanying metadata scores the availability and reliability of each category of input source by country, unsurprisingly indicating that the input mix of these sources varies substantially by country. Notably for the percentile-level panels, the income data exhibits substantial bottom-coding with almost every country-year reporting exactly zero income for at least the bottom four percentiles of the pre-tax income distribution. Even if an accurate measure of adult unemployment, this evokes well-known deficiences of income as an identifier and as measure of poverty compared to alternatives like consumption expenditures (Meyer and Sullivan, 2003). For example, while the unemployed subpopulation of a given economy may well be composed of the most impoverished, it is likely to also include individuals of all levels of precarity and social status and does not account for security provided by communal and social safety nets. It also positively biases estimated changes in welfare among the poor since zero incomes mechanically cannot decrease.

Given these inherent disadvantages of income measures for representing poverty, we intentionally do not make specific claims about the poverty implications of our findings even though this is obviously a first-order distributional concern. Instead, we aggregate the incomes of the bottom 10 percentiles in all analyses to consider the bottom 10% of the income distributions a single grouping and consider estimated impacts for this quantile to be a conservative bound on true impacts on the very poorest.

Climate data

We merge all our panels with a panel of annual population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). This data corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Department of Civil and Environmental Engineering/Princeton University, 2006). The GMFD reports historical weather data for the entire global surface at 0.25-degree resolution[8] for every three hours

[8] At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

from the start of January 1, 1948 to the end of December 31, 2015.

2.3 Distributional effects of temperature shocks

The first set of distributional results we present proceeds similarly to the expositional GDP regressions in 2.1 but with separate implementations for each within-country quantile q of interest. For every within-country-quantile group $q \in Q$, we estimate the $H+1$ state-dependent local-projection regressions for horizons $h \in \{0, 1, ..., H\}$ described by the following single-equation model:

$$\Delta_h^c y_{i,t+h}^q = m_{1,h}^q \widehat{\tau}_{i,t+h}^c + m_{2,h}^q \widehat{\tau}_{i,t+h}^c \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}^q \boldsymbol{\gamma}_h^q + \mu_i + \eta_t + \varepsilon_{i,t+h}^q \qquad (4)$$

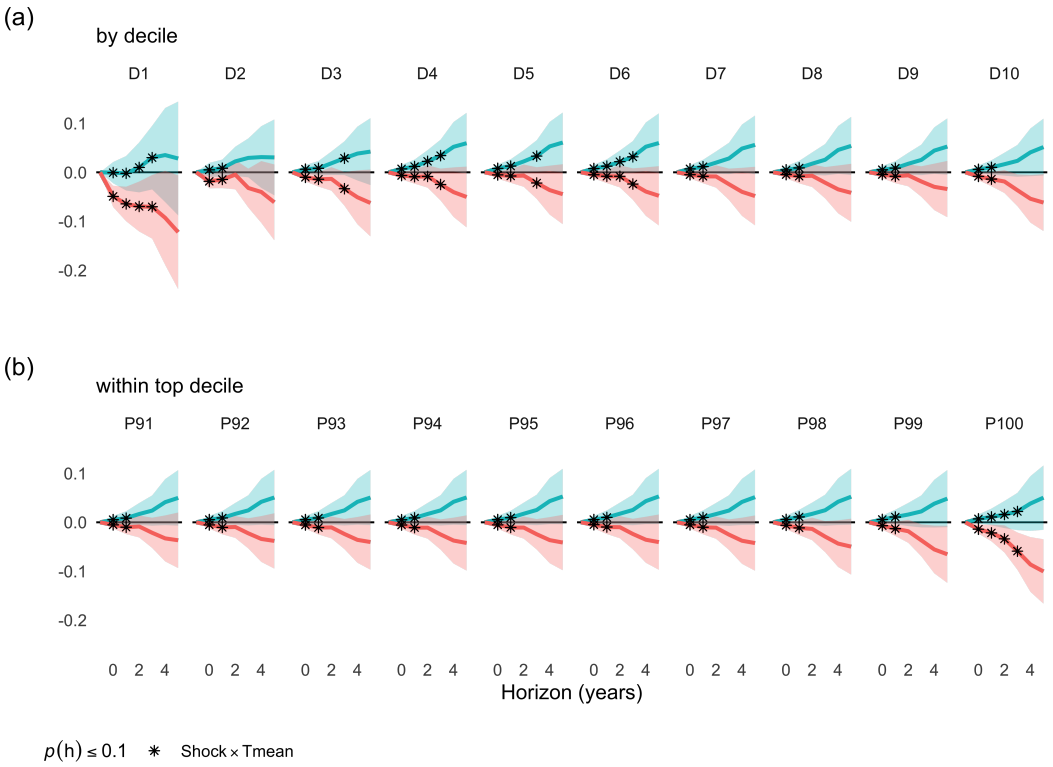
Here, variables denoted by a superscript c are cumulative variables constructed using the one-step process described in A1 to calculate cumulative effects which account for the persistence of temperature shocks. The quantile-specific matrix of controls $\mathbf{Z}_{it}^q$ include lags $\ell \in \{1, ..., L\}$ of the shocks, the interaction of the shock with the state variable, the outcome $\Delta y_{i,t-\ell}^q$, and country-level aggregate growth $\Delta y_{i,t-\ell}$. Multiplier stimates $\{\widehat{m}_{1,h} + \widehat{m}_{2,h}\}_{h=0}^H \overline{T}_i$ describe the effect of an identified temperature shock on the cumulative stock of outcome y^q after h periods as a function of the expected temperature.

For example, panel (a) of Figure 5 summarizes the results from estimating separate sets of cumulative temperature multiplier functions for each of 10 deciles. Here, color-coding follows the same intuitive scheme as previous plots with blue and red corresponding to representative expected temperatures of 10 and 26°C respectively[9]. The points in asterisks indicate significant interactions which impose differentiated effects based on expected climate. These effects persist for between 1-3 years, concentrating negative impacts in hot countries while keeping cooler countries relatively insulated, consistent with the aggregate results standard in the climate literature and the dynamic results reported in Section 2.1 using GDP. Estimation error is larger than when estimating GDP impacts in Figure 4, unsurprising given the difference in temporal coverage and data quality.

New here is the ability to compare impacts across quantiles within countries. Focusing on hot-climate countries, the estimates reported in Panel (a) imply an incidence pattern wherein negative effects of a transitory temperature shock to hot countries are found at all quantiles but are particularly dire

[9]We omit the green 18°C main effect to avoid over-busy plots, but it is of course implied as the average of the two depicted curves

Figure 5: **Cumulative state-dependent income response to a 1°C shock**



at the bottom. For 26°C countries, the 90% confidence intervals for the impact of a temperature shock on cumulative income a year after the shock is $[-10.0, -2.9]$ percent for the bottom decile compared to $[-3.2, 0.3]$% for the second decile. This generally improves as one moves up the income distribution, stabilizing at approximately $[-2.4, 0.6]$ percent by the fourth decile through to the 99th percentile. Interestingly, the confidence interval then discontinuously drops to $[-3.9, -0.5]$ for the richest percentile. Panel (b) shows this discontinuity within the top decile and also indicates this effect persists for two additional years. In contrast, impacts are roughly stable for cold countries. For the coolest countries, cumulative growth incidence effects are much flatter. Point estimates suggest positive impacts are somewhat larger between the third and sixth deciles.

Figure 6a and the bottom panel of Figure 6b capture these discontinuities at the extremes more clearly by plotting point estimates of percentile-level estimated over time for our three representative climates. The implication then is that negative impacts to hot countries are concentrated on both the bottom 10% and the top 1%[10]. To the best of my knowledge, this particular vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously documented or theorized. In ongoing work, I investigate the extent to which this may be driven by weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of weather derivatives in these settings.

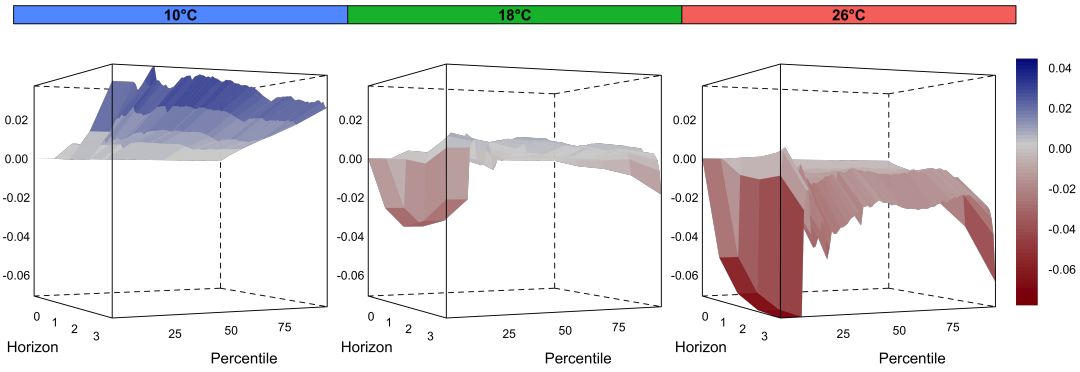
This first set of new results describe inequalities in the growth incidence of impacts across a national income distribution as a function of the local climate. The disproportionate impacts to the bottom decile strongly imply a regressive effect: In the aftermath of an isolated temperature shock, a representative member of the the bottom decile in a 26°C country can expect to earn 7% less over the next three years than they would had the shock never occurred. This compares to losses of 2.2% for the median adult in their country and a *gain* of 3.0% for their counterpart in a country with a 10°C climate. At the same time, a representative member of the top 1% in the same 26°C country, can expect to earn 5.9% less over those three years than they would otherwise compared to 3.6% for the 99th percentile. Other than the

While these results strongly imply stark a redistributive effect of a temperature shock, their implied effect on inequality depends both on the initial

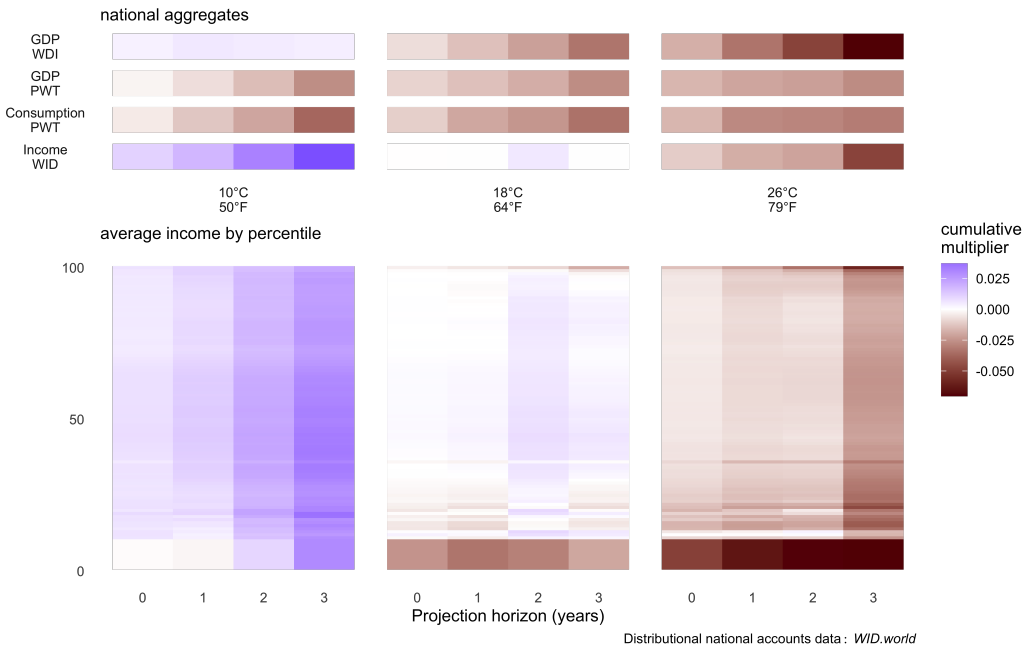
[10] As mentioned in Section 2.2, the income data of the bottom 10% is not well suited for decomposition into percentile effects but it would be natural to expect that impacts to the bottom decile would produce a similarly steep within-decile gradient in which negative impacts are driven by the poorest percentiles within the bottom 10%.

Figure 6: **Point estimates of cumulative income growth incidence responses to a 1°C shock**

(a) Incidence patterns for three representative climates

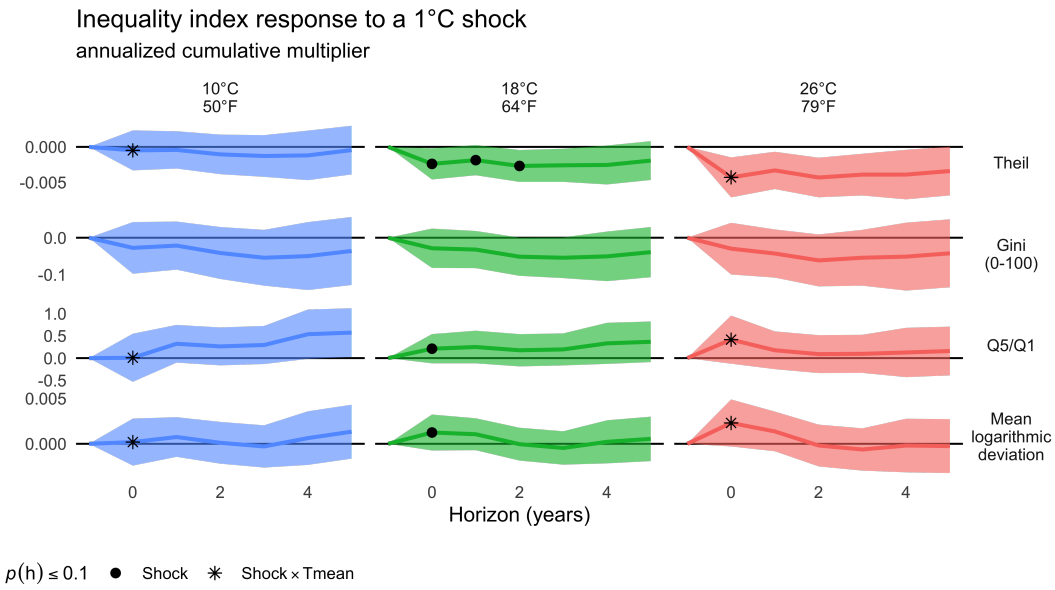


(b) Aggregate vs. distributional responses



distribution of income and the measure of choice, particularly given its non-monotonic features. Figure 7 depicts the annualized effect of a temperature shock on various measures of inequality. Units are not directly comparable across measures but all measures are increasing in (their respective conception of) inequality. Measures are provided in order from the most sensitive to redistribution of top incomes (the Theil index) to the most sensitive to redistribution of bottom incomes (the mean logarithmic deviation index).

Figure 7: **Inequality index response to a 1°C shock**



The filled circles in the middle column drawn with green error bands correspond to significant main effects of the temperature shock on the given measure of inequality. The asterisks on the plots on either side correspond to significant and persistent interactions, measuring linear heterogeneity by expected temperature. In light of the growth incidence results just discussed, the near-zero impacts to cold-country inequality and substantial effects to hot-country inequality are unsurprising. Null effects on the Gini index, known to be relatively insensitive to changes at either end of the income distribution, are consistent with the disproportionate impacts on the bottom 10% and top 1%.

By the Theil index, a transitory temperature shock appears to improve in-

equality while inequality as measured by the ratio of incomes of the top 20% to the bottom 20% and the mean logarithmic deviation index both suggest the opposite. From the previous analysis, we can surmise that the progressive effect on the Theil index is being driven by the transfer away from top 1% while the regressive transfer of income shares away from the bottom 10% outweighs this consideration for the other two measures. While the impacts to the bottom 10% are substantially larger and represent 10 times as many individuals as the impact to the top 1%, the disproportionately large share of income held by this elite minority constitutes a large enough share of total income that these measures disagree on the overall effect.

3. Implications of climate change for global welfare

In this section, we extend the empirical within-country analysis of Section 2 to an analysis of global inequality. The variety of reduced-form within-country effects estimated thus far do not on their own identify an overall effect on global inequality. In our framework, this will depend not only on the relative total incomes and population sizes of the countries which make up the world, but also their climates(which moderate the effect of the shock in (4)) and the spatial distribution of temperature shocks, which we observed varied substantially over time and across different climates in Figure 1).

Finally, to attribute an inequality effect to historical anthropogenic climate change, we implement a counterfactual analysis which can decompose observed temperature effects into their natural and anthropogenic origins, adapting methods commonly used in climate science. We describe the method of performing this analysis in Section 3.2 then present results in Section 3.3.

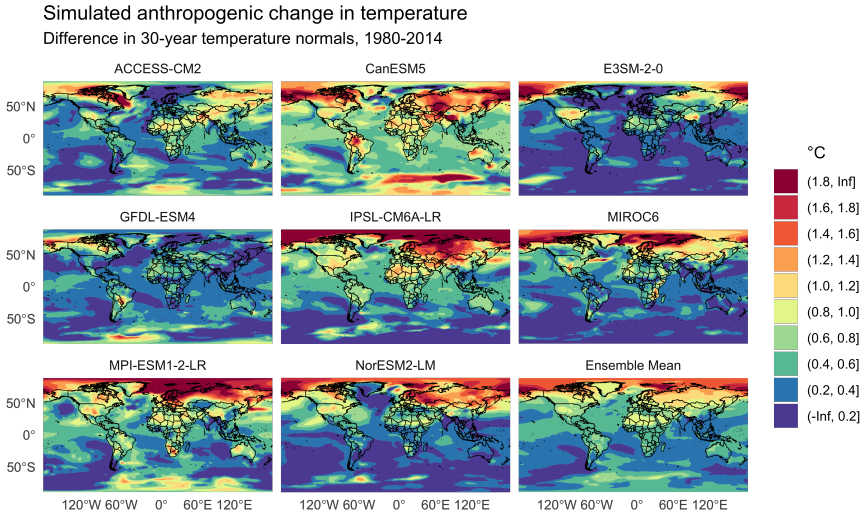
3.1 Implications for global inequality

3.2 Historical counterfactuals

3.2.1 Climate simulation data

Climate simulation data come from the contributions of climate modeling groups to the World Climate Research Program's Coupled Model Intercomparison Project (CMIP). The CMIP, initiated in 1995, is a framework to coordinate, standardize, and disseminate the results of state-of-the-art simulations of the global climate for the benefit of the international climate research com-

Figure 8: **Model-implied spatial distributions of global warming, 1980-2014**



Simulated spatial distribution of anthropogenic warming for eight randomly selected climate models. The bottom-right plots the average distribution for all 13 CMIP6 models which ran both historical and historical-natural simulations.

munity. Data used in this study comes from CMIP6, the sixth and newest[11] generation of models (Eyring et al., 2016). Data is provided as a collection of globally gridded time series summarizing the joint evolution of several hundred climate variables under various pre-specified calibrations. We will primarily be interested in the models' simulations of near-surface air temperature.

All participating models are "coupled", meaning that they explicitly account for interactions and feedback between distinct components of the global climate system such as the atmosphere, the cryosphere, land surface, and the ocean. Because the manner in which this integration is achieved is idiosyncratic to each model, computationally expensive, and highly sensitive to external calibrations, coordinating the set of "experiments" that participating models run is necessary for comparability and interpretability. For example, one of the four indexes which define an experiment is the realization index which corresponds to the set of geophysical initial conditions at the beginning of the simulation period. By holding the initial conditions fixed, differences in results across models within the same experiment can be more readily attributed to the distinguishing features of the model without being confounded by differences in implementation. Similarly, differences in the same climate model's results across different perturbations to the initial conditions provide a measure of the model's internal variability. The CMIP thus provides a systematic way of decomposing total variation across models and experiments both internal and external to the model. Coordination also enables identification of systematic and idiosyncratic biases in model designs that then inform the next generation of models.

Among the other three indexes which define an experiment, the forcing index is central to our counterfactual analysis. Forcings are factors which affect the net transfer of radiative energy in the climate system, essentially the energy the Earth receives from the sun minus that which it expels back into outer space. Anthropogenic forcings include, for example, industrial greenhouse gas emissions, deforestation, and the transformation of land for arable agriculture. Natural forcings include aerosol expulsions from major volcanic activity and variation in solar irradiance such as those associated with the 11-year solar cycle.[12] Comparing simulations which only differ in their forcings allow one to attribute differences in outcomes to the differences in forcings; we describe next how comparisons of this kind form the basis of both our historical and projection counterfactual analyses.

[11] After a delayed rollout, outputs from most CMIP6 models were made accessible by 2022

[12] Natural forcings are distinct from sources of natural variability such as the ENSO cycle because they are external to coupled climate models.

Construction of historical counterfactuals

Our historical counterfactual analysis uses simulations from a category of experiments called "historical runs". For the CMIP6 generation of models, historical simulations begin at a pre-industrial baseline period in 1850 and run through to 2014. Among the forcing scenarios used in these historical experiments is one simply labeled "historical", which attempts to represent the impacts of influential forcings actually imposed on the climate system over this period, such as the deforestation of the Amazon rainforest and the eruptions of Krakatoa in 1883 and Mt. Pinatubo in 1991. Results from this historical run are required of all participating CMIP6 models since these "hindcasts" can be directly compared to observational data in order to evaluate model accuracy. Models collectively perform remarkably well at reproducing the historical record (Zhou et al., 2020; Zelinka et al., 2020; Liu et al., 2021). For this reason, studies which use CMIP data for the purpose of counterfactual analysis are advised to include the full available "multi-model ensemble" for experiments of interest (Tebaldi and Knutti, 2007; Knutti et al., 2010; van Vuuren et al., 2006).

A subset of these models also simulate a set of optional 'historical-natural' experiments which are defined identically except with anthropogenic forcings held fixed at pre-industrial levels. The differences between the two forcings by the same model holding all other variables fixed is then interpreted as the model's simulation of anthropogenic contributions to climate change. Comparisons between historical and historical-natural experiments are commonly used in the subfield of attribution science concerned with quantitatively measuring anthropogenic contributions to the intensity, frequency, or probability of weather events or trends.

For each climate model m and country i, we use an observed series T and the two model-specific historical series $\mathcal{T}^{hist}$ and $\mathcal{T}^{nat}$ to construct a counterfactual series $\widetilde{T}$ where anthropogenic forcings have been held fixed since the first year of the simulation period. The iterative process is summarized in Algorithm A2 and illustrated in Figure 9 using data from the Philippines as an example. In panel (a), differences between model-specific simulations of national temperature when including all forcings (orange) and when excluding anthropogenic forcings (green horizontal axis). Local regression fits are depicted in red. Light-blue shaded regions correspond to 30-year reference periods used to define temperature normals (dark blue dots) for the start and end of the simulation period 1980-2014.

In panel (b), The linearized difference between normals depicted as dark blue lines in (a) are subtracted from the observed temperature series (black)

starting in 1980 to produce counterfactual temperature series (green). Each of these is interpreted as the temperature history that would have been realized had anthropogenic forcings been held fixed at 1980 levels according to a specific model of the climate system.

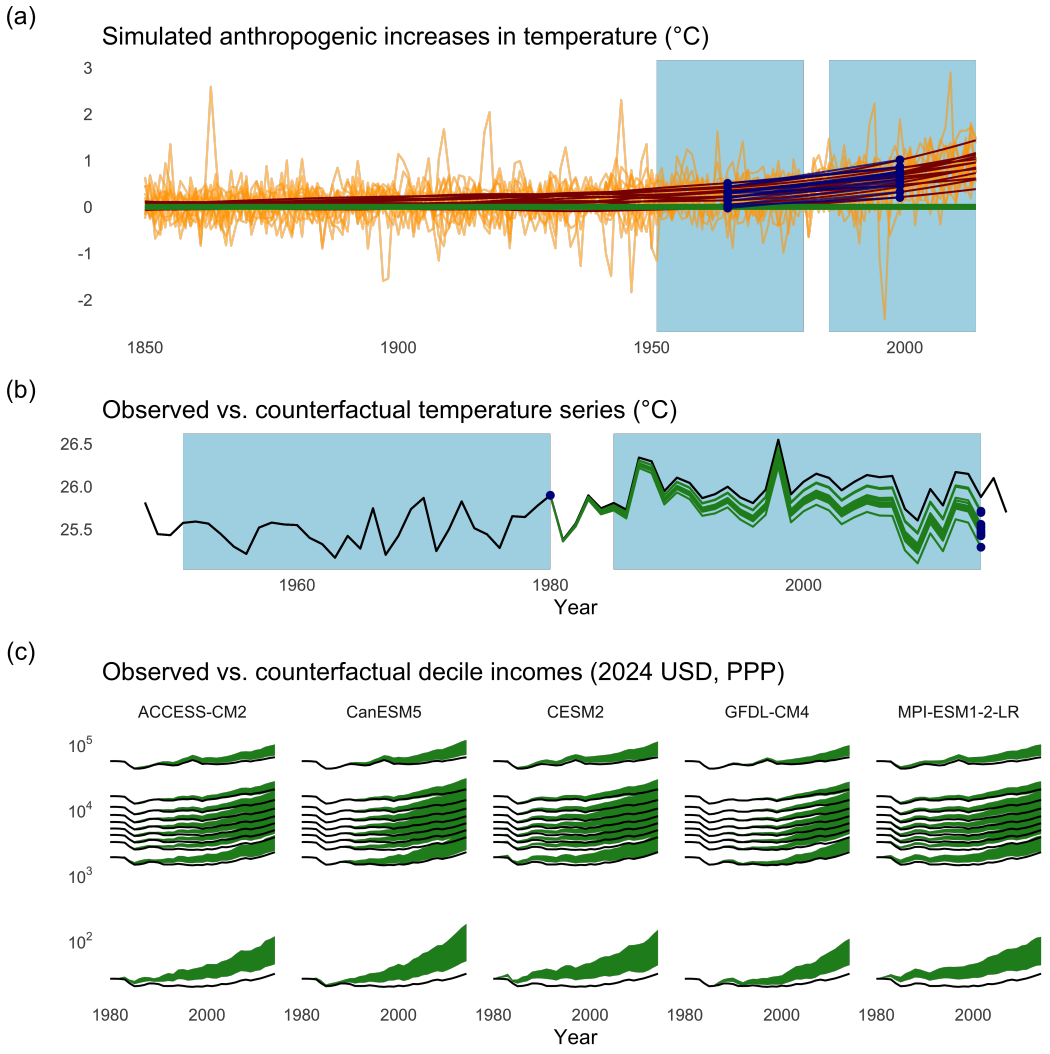
In panel (c), linearizing the difference between the climate normals defined at either end of the simulation period and subtracting the result from an observed temperature series implements the standard "delta method" for constructing bias-corrected climate counterfactuals. Counterfactual economic series for a random selection of five climate models are depicted in green and are constructed by iteratively applying the decile-level dynamic income responses depicted in Figure 5 to the observed and counterfactual temperatures depicted in (b). Their difference is interpreted as the change in income attributable to anthropogenic activity since 1980. Green bands correspond to 90% confidence intervals generated from 500 bootstrap estimates of the response functions.

3.3 Counterfactual results

Anthropogenic contributions to observed inequality since 1980

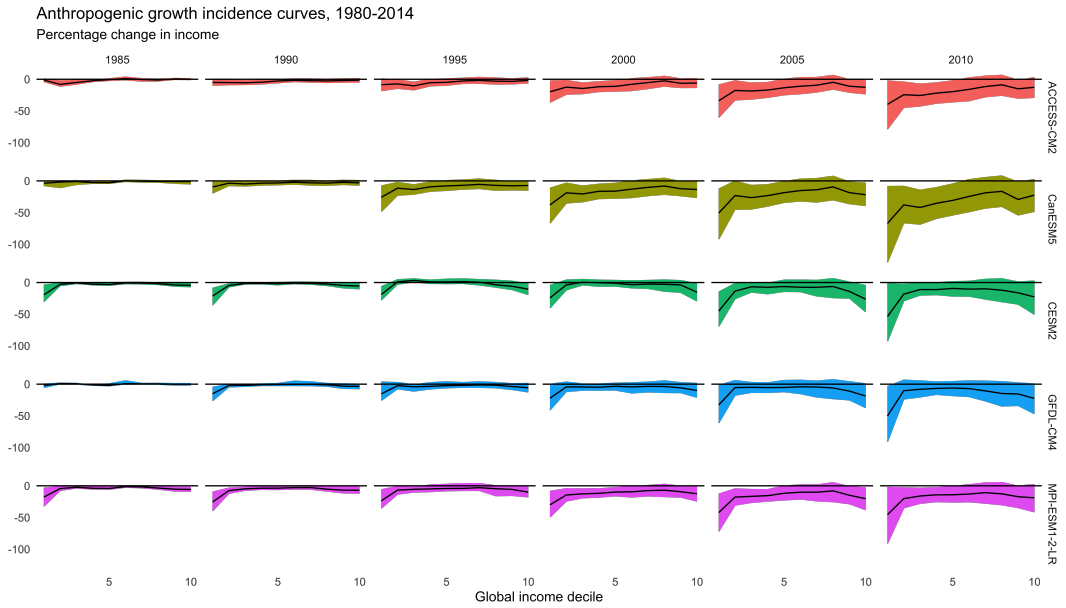
Global inequality decomposition: within-country vs. between-country

Figure 9: **Constructing historical counterfactuals holding anthropogenic forcings fixed at 1980 levels**



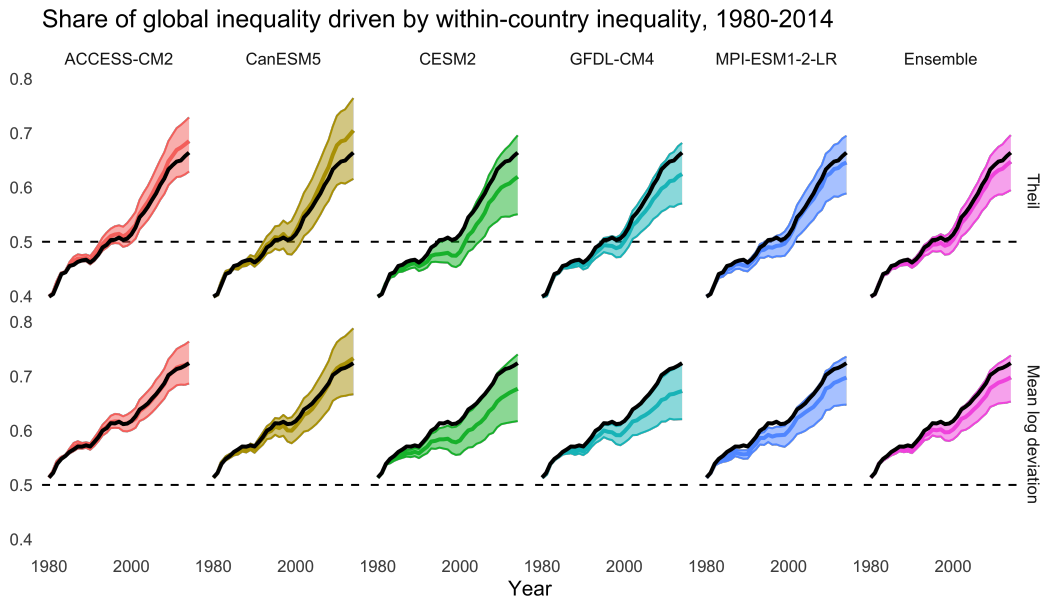
Demonstration of the delta method for constructing bias-corrected counterfactuals using data for the Philippines as an example.

Figure 10: **Anthropogenically attributable change in income by global income percentile**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

Figure 11: **Within-country and between-country inequality, observed and counterfactual**



4. Discussion

To the best of my knowledge, this particular vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously documented or theorized. In ongoing work, I investigate the extent to which this may be driven by weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of weather derivatives.

It is worth taking seriously criticisms that a "profound indifference to inequality" (Sen et al., 2020) nonetheless remains embedded in the varieties of cost-benefit analyses predominant in the climate economic literature. The concavity of the chosen utility function does not change the underlying utilitarian social calculus which has long been subject to claims it "does not take seriously the distinction between persons" (Rawls, 1971; Sen, 1999). This evokes a broader "disappearance of welfare economics" lamented by senior scholars of poverty and inequality (Atkinson, 2001, 2011; Sen et al., 2020)[13]. For example, the progressive weighting scheme intended to reflect greater consideration for those most impacted by climate change directly prescribes carbon price increases which may disproportionately burden the segments of the population least willing or able to afford them. The historical unpopularity of carbon pricing measures historically might be read as a consequence of this legacy of disengagement.

4.1 Conclusion

[13]The historian Eli Cook provides provocative intellectual histories (Cook, 2019, 2022)

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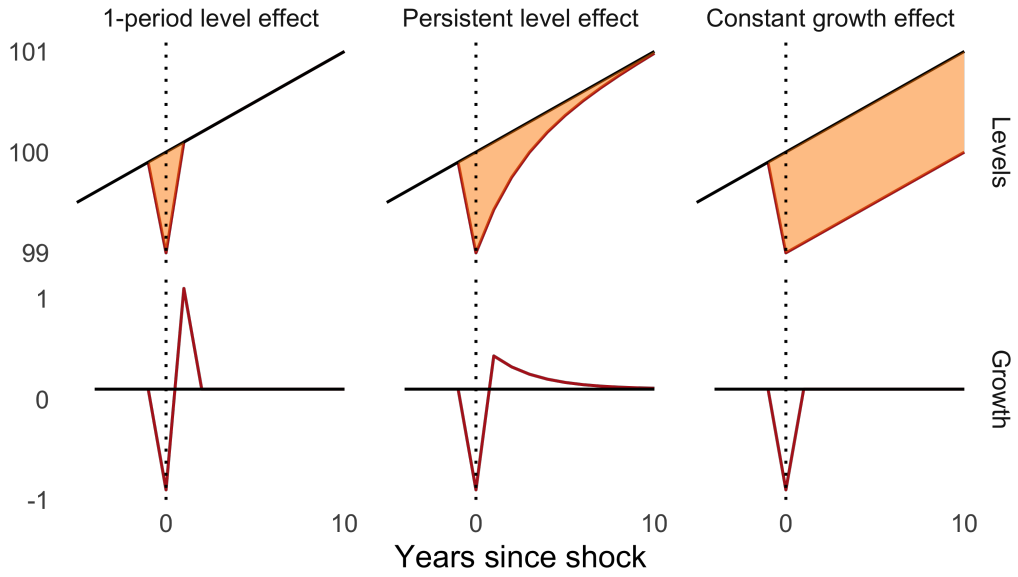
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A1. Empirical methods

A1.1 Growth vs. level effects

Figure A1: **Levels and growth under different persistence structures**



Each column corresponds to a different possible dynamic path of a temperature shock in period 0 on an outcome measured in levels (top panels) and annual growth (bottom). Black and red lines depict the evolution of the outcome variables in the absence and presence of a unit shock in period 0 respectively under the different described dynamics. Filled areas in orange measure the cumulative difference in the outcome attributable to the shock. In the context of climate change, these represent the long-term damages associated with a temperature shock.

Under a one-period level effect, the level of, say, GDP falls by an estimated level $\widehat{beta}_0$ iin the year of the shock but fully recovers to trend in the next period. All lost production is compensated for in the next period by virtue of an above-trend rate of growth which exactly offsets the negative growth effect in the year of the shock. The shock has no effect on future production and the area in orange represents a one-off economic loss from the shock.

Under the constant growth effect depicted in the rightmost panels, the same shock incurs the same effect on the levels but this effect is permanent so that GDP in all future periods is reduced by the same amount relative to the no-shock counterfactual. The difference between the two trajectories thus compound over time as observed by the orange area which grows linearly over time. The growth effect depicted in the bottom panel is entirely transitory.

The middle column describes the intermediate case where the economy requires several periods to return to its pre-shock trend. Cumulative differences attributable to the one-off shock continue to rise over time but to a diminishing degree until eventually compensatory positive growth exactly offsets the initial negative growth, i.e., the integral of the growth curve equals zero in the long run.

Now consider the following two models, which we may refer to as the levels and growth models respectively.

$$\log Y_t = \beta^L X_t + v_t \tag{5a}$$

$$\Delta Y_t = \beta^G X_t + u_t \tag{5b}$$

In the levels model, β^L represents the difference between the observed level of GDP and the GDP that would have occurred had a unit shock in period 0 not occurred. Graphically, it is the vertical distance between the black and red lines in the levels panels. Since this is an entirely static model, there is no mechansim for X to affect future values of Y and so the model assumes a perfect return to trend in all ensuing periods. In the growth plots, this is represented by a $\beta^L - 0 = \beta^L$ effect in period 0 and an exactly offsetting $0 - \beta^L = -\beta^L$ effect on growth in period 1 and no effect on growth in all ensuing periods. Level effects which persist for p periods can be accommodated by including p additional lags of X in the model. A permanent growth effect can only be accommodated with the inclusion of infinite lags.

In the growth model, $\beta^G \neq 0$ represents a permanent change in the difference between consecutive values of Y_t; the gap between the black pre-shock trend line and the observed GDP series stays fixed at β^G for all periods following the shock and is never offset. Level effects which persist for p periods can be accommodated by including p lags of X and finding that the sum of the $p + 1$ coefficients is exactly 0

A1.2 Inferring persistence using significance bands

Figure A2: **GDP impulse response to a 1°C shock, including significance bands**

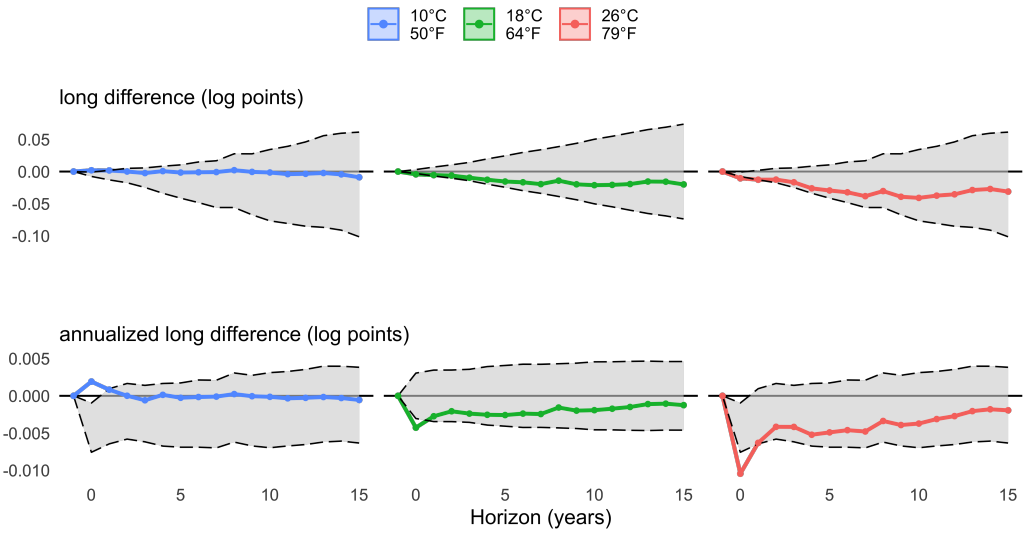
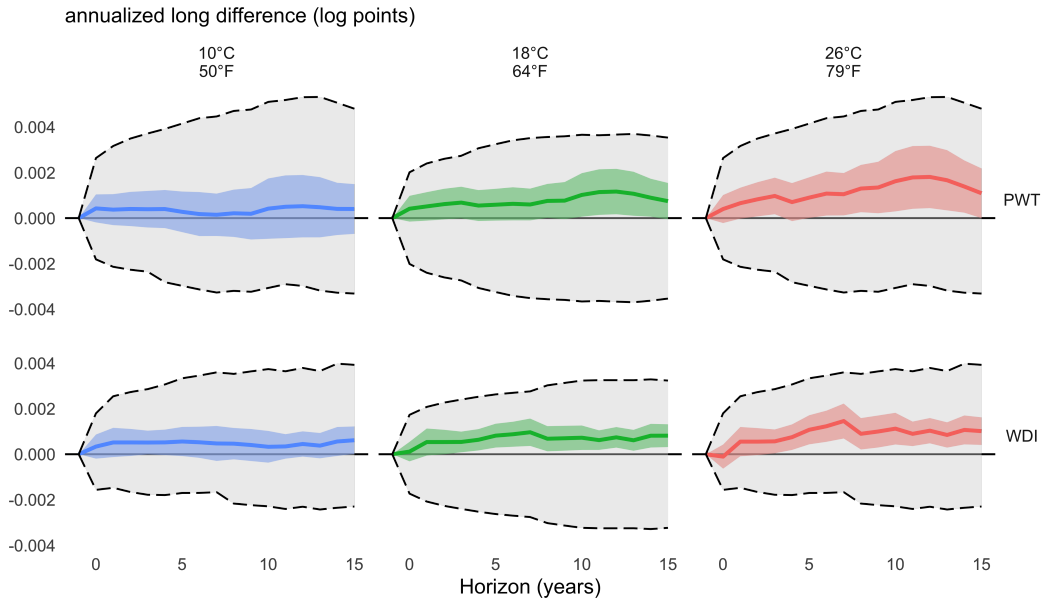


Figure A3: **Placebo test: Estimating national population impulse response to a 1°C shock**



A1.3 Data processing

Algorithm A1: One-step state-dependent cumulative multiplier

for $h = 0$ **to** H **do**

Estimate the IRF $\mathcal{R}_{\tau\to\tau}(h)$ as defined in (3), collecting state-dependent shock coefficients $\widehat{\alpha}_{1,h}, \widehat{\alpha}_{2,h}$;

Define $\Delta_h^c y_{i,t+h} := \sum_{j=0}^{h} \Delta_j y_{i,t+j}$;

Define $\widehat{\tau}^c_{i,t+h} := \widehat{\tau}_{it}\left[\sum_{j=0}^{h} \widehat{\alpha}_{1,j} + \widehat{\alpha}_{2,j}\overline{T}_{it}\right]$;

Estimate local projection:

$\Delta_h^c y_{i,t+h} = m_{1,h}\widehat{\tau}^c_{i,t+h} + m_{2,h}\widehat{\tau}^c_{i,t+h}\overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h}$;

end

Output: Multiplier coefficients $\widehat{m}_{1,h}, \widehat{m}_{2,h}, \widehat{\lambda}_h$ for $h \in \{0, ...H\}$

Algorithm A2: Constructing temperature counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\mathcal{T}^{hist}_{m,i,t}\}$,$\{\mathcal{T}^{nat}_{m,i,t}\}$,$\{T_{it}\}$

for $t^* \in \{1980, 2014\}$ **do**

$\overline{\mathcal{T}}^{hist}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{hist}_{m,i,t^*-j}$;

$\overline{\mathcal{T}}^{nat}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{nat}_{m,i,t^*-j}$;

$\overline{\delta}_{m,i,t^*} \leftarrow \overline{\mathcal{T}}^{hist}_{m,i,t^*} - \overline{\mathcal{T}}^{nat}_{m,i,t^*}$;

end

for $t = 1980$ **to** 2014 **do**

$\overline{\delta}_{m,i,t} \leftarrow (t - t_0)\frac{\overline{\delta}_{m,i,2014} - \overline{\delta}_{m,i,1980}}{2014-1980}$;

$\widetilde{T}^{nat}_{m,i,t} \leftarrow T_{it} - \overline{\delta}_{m,i,t}$;

end

Construct states $\{\overline{T}^{nat}_{m,i,t}\}$ by applying (1a) to $\{\widetilde{T}^{nat}_{m,i,t}\}$ then subtracting by $\widetilde{T}^{nat}_{m,i,1980}$;

Construct shocks $\{\widehat{\tau}^{nat}_{m,i,t}\}$ by applying (1a) and (1b) to $\{\mathcal{T}^{nat}_{m,i,t}\}$;

Output: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$

Algorithm A3: Constructing economic counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$, B bootstrap samples indexed b

Initializing series:

  for $b = 1$ **to** B **do**

    Specify desired simulation interval: $(t_0, t_1) \leftarrow (1980, 2014)$;

    Estimate $\mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b)$ from (4);

    Specify maximum projection horizon H using persistence test from Section 2.1.2;

    Define IRFs $\mathcal{R}_{\tau \rightarrow Y}(h; \overline{T}, b) := \mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b) - \mathcal{R}^{c}_{\tau \rightarrow Y}(h-1; \overline{T}, b)$;

    Collect IRF coefficients;

    $\{\widehat{\beta}_{1,h,b}, \widehat{\beta}_{2,h,b}, \widehat{\lambda}_{h,b}\}_{h=0}^{H}$;

    Define $\delta^{\tau}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} - \widehat{\tau}_{it}$;

    Define $\delta^{\tau T}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} \cdot \overline{T}^{nat} - \widehat{\tau}_{it} \cdot \overline{T}_{it}$;

    Define $\delta^{T}_{m,b,i,t} := \overline{T}^{nat}_{m,i,t} - \overline{T}_{it}$;

    Initialize $\delta^{Y}_{m,b,i,t} \leftarrow 0$;

    Initialize $\widetilde{Y}^{nat}_{m,b,i,t_0} := Y_{i,t_0}$

  end

Iteratively constructing series:

  for $t = t_0 + 1$ **to** t_1 **do**

    for $b = 1$ **to** B **do**

      for $h = 0$ **to** H **do**

        $\delta^{Y}_{m,b,i,t+h} \leftarrow \delta^{Y}_{m,b,i,t+h} + \widehat{\beta}_{1,h,b} \cdot \delta^{\tau}_{m,b,i,t} + \widehat{\beta}_{2,h,b} \cdot \delta^{\tau T}_{m,b,i,t} + \widehat{\lambda}_{h,b} \cdot \delta^{T}_{m,b,i,t}$;

      end

      $\widetilde{Y}^{nat}_{m,b,i,t} \leftarrow \left(1 + \log(\frac{Y_{it}}{Y_{i,t-1}}) + \delta^{Y}_{m,b,i,t}\right) \widetilde{Y}^{nat}_{m,b,i,t-1}$

    end

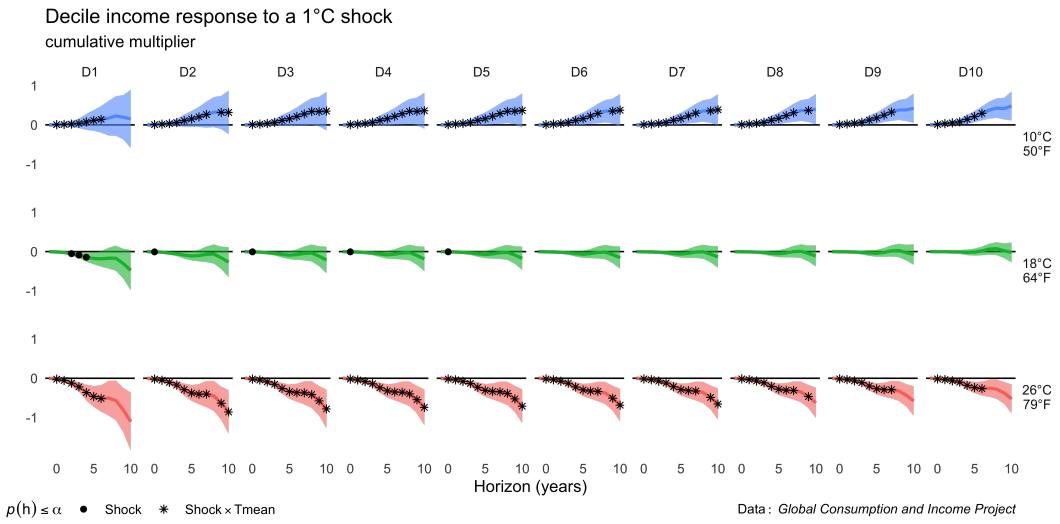
  end

Output: $\widetilde{N}^{nat}_{m,b,i,t}$

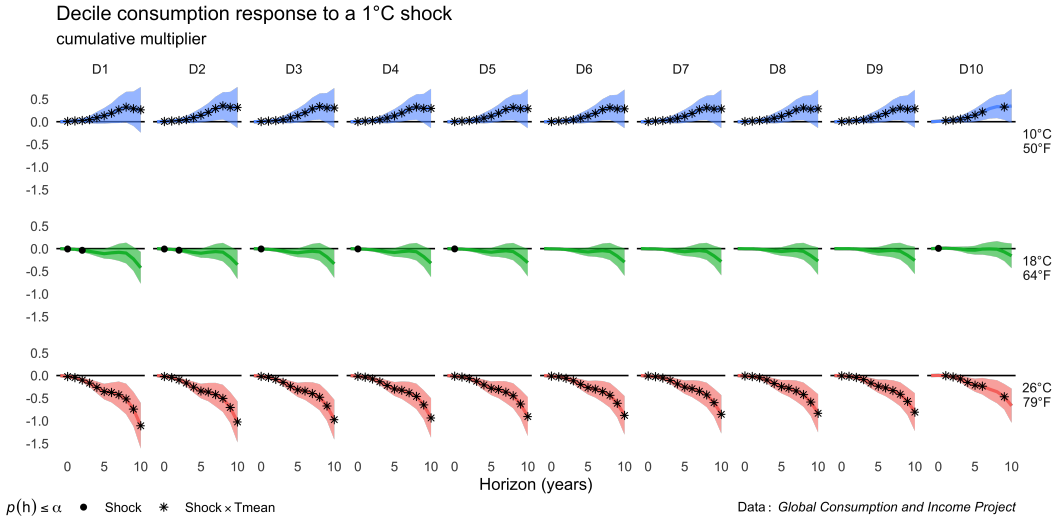
A2. Supplementary empirical results

Figure A4: **Income and consumption multipliers by decile, estimated using survey microdata**

(a)

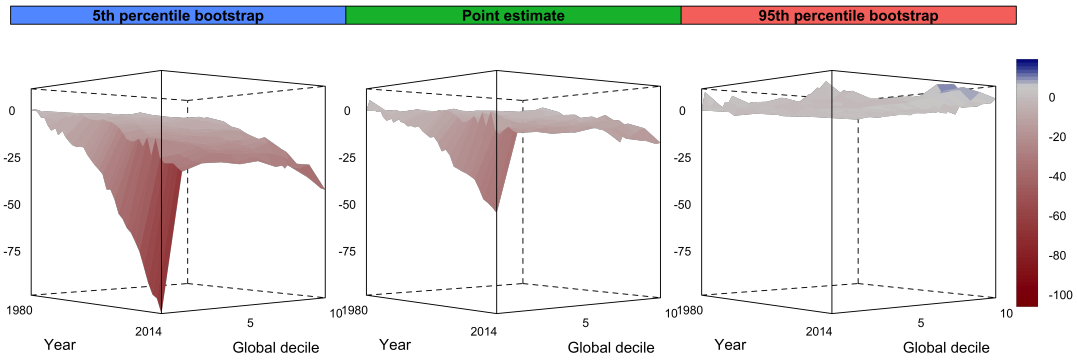


(b)



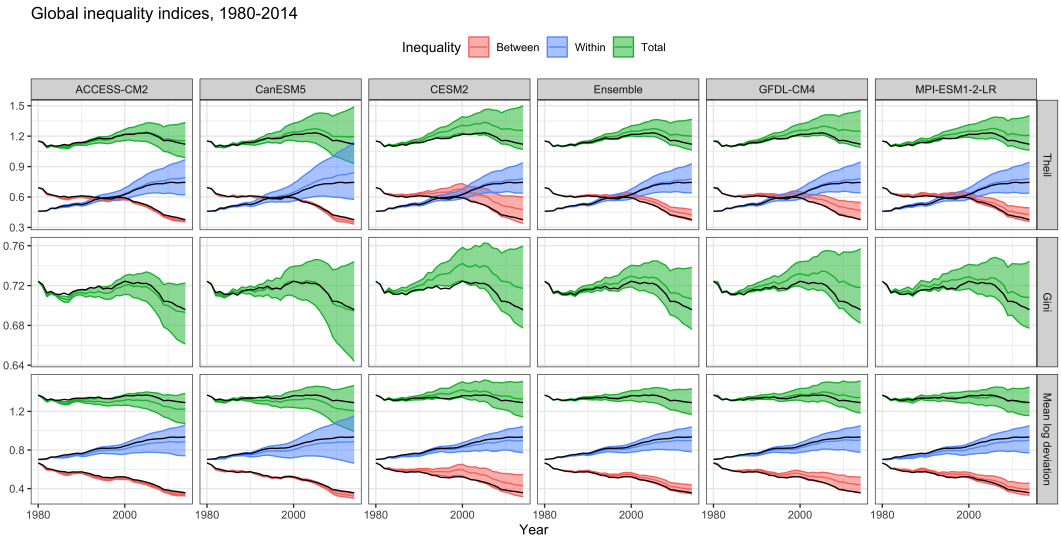
A3. Supplementary simulation results

Figure A5: **Anthropogenically attributable change in income by global income percentile**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

Figure A6: **Measures of global inequality, observed and counterfactual**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.